Shopping Customers

Segmentation and Clustering

Problem Statement: Understand the ***Target Customers*** for the marketing team to plan a strategy that will increase sales and improve customer engagement through personalized marketing campaigns to improve conversion rates, attract, engage, and retain the **most profitable customer segments.**

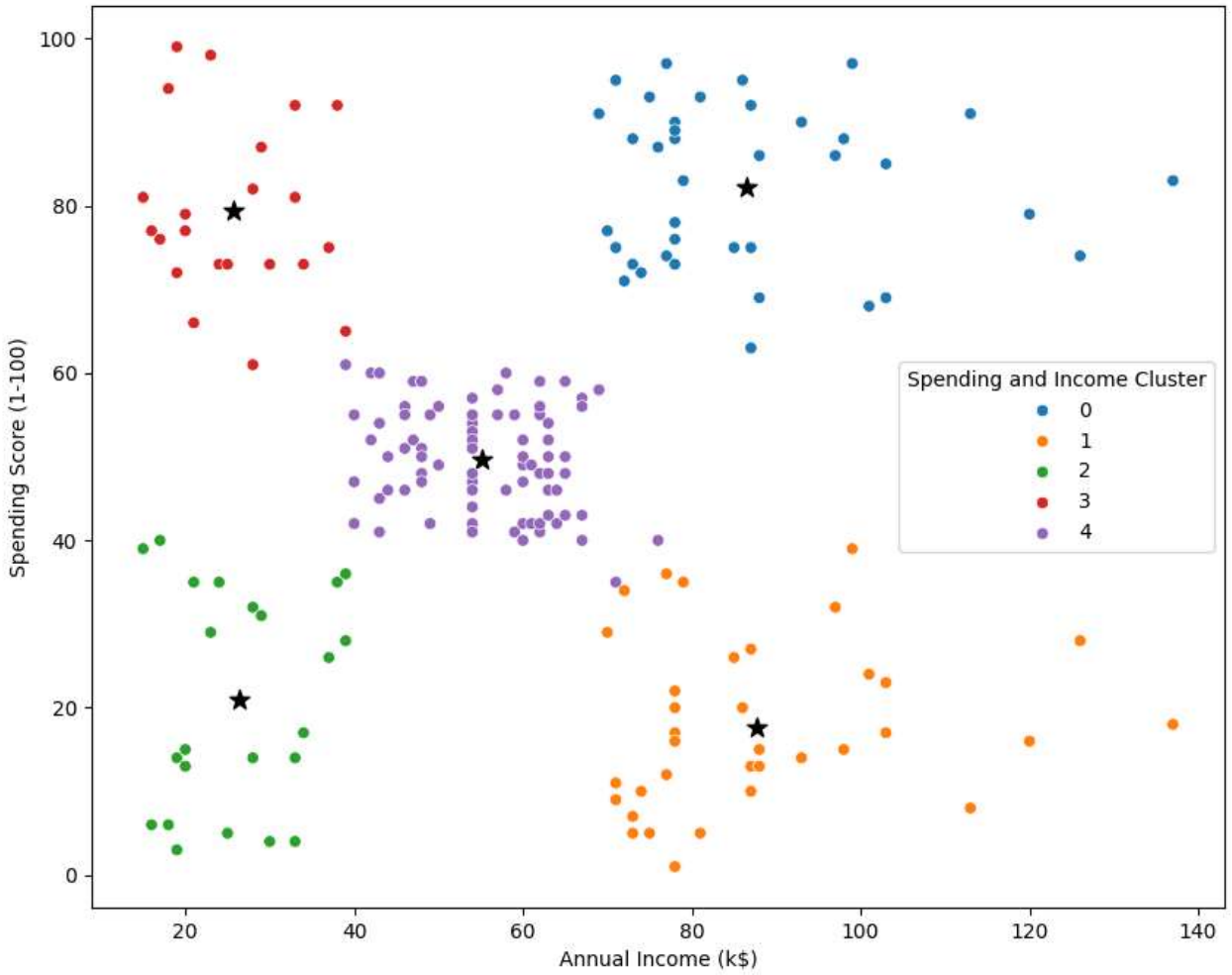
Context: Identify the ***most important shopping groups*** based on ***income, age***, and the mall ***shopping score***. Produce an ideal number of ***groups*** with a label for each.

Approach:

- Perform explanatory data analysis (EDA).
- Use KMeans Clustering Algorithm (Kmeans from Sklearn) to create customer segments based on spending and income.
- Use Summary Statistics on the clusters.
- Visualize.

Target Cluster Findings and Recommendations:

- The primary target group is ***Cluster 0,*** as it has both a ***high spending score and high income,*** making it the ***most valuable customer segment.***

- Approximately ***60% of the shoppers in Cluster 0 are women***. Marketing efforts should focus on ***attracting these customers*** through campaigns that highlight ***popular and trending products*** within this segment.

- ***Cluster 3*** also presents a promising opportunity through their high spending score despite their lower income. Customers in this group could be targeted with ***promotions and sales events***, especially for popular items, to encourage increased spending.